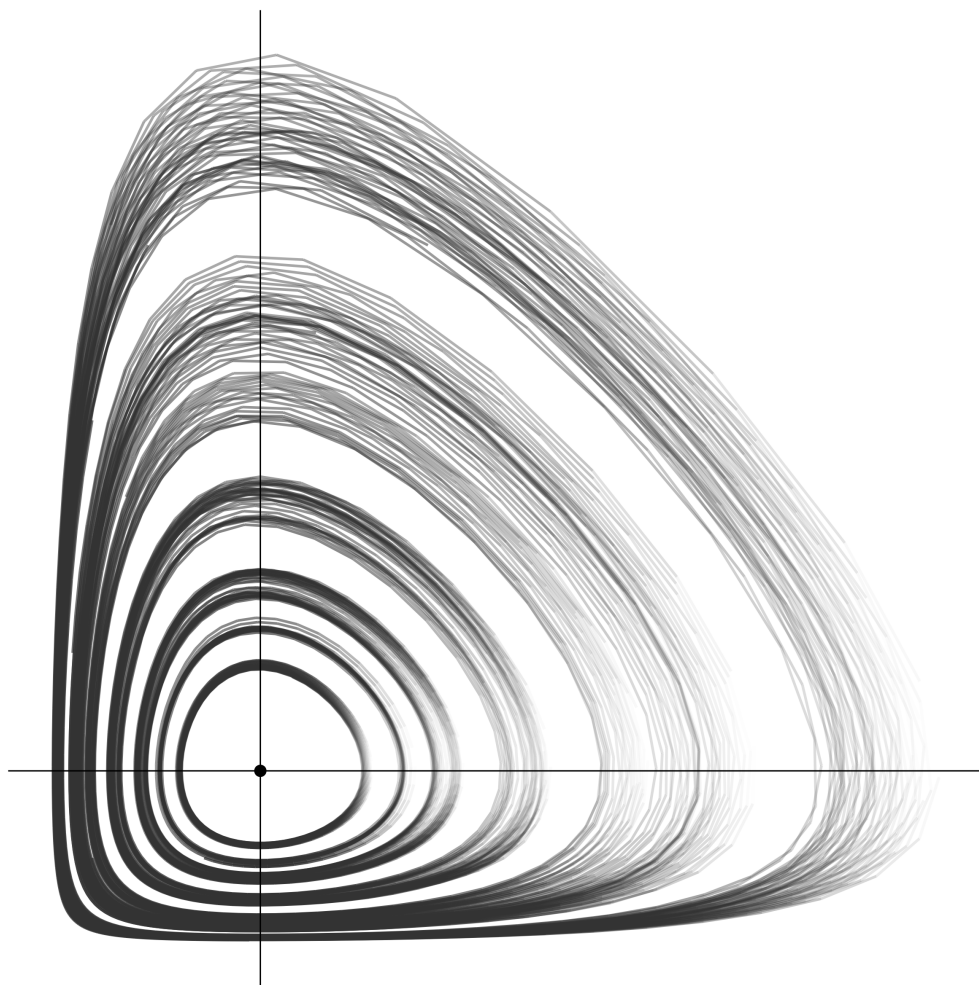

AMATH 351

Ordinary Differential Equations II

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Credit to Prof. Mohsen Rezaeian for His Lectures on AMATH 351

Preface

A majority of the information presented in this note package is transcribed directly from the *AMATH 351: Ordinary Differential Equations II* lectures and the provided course notes during the Spring 2025 term, taught by Prof. Mohsen Rezaeian.

These notes are by no means exhaustive nor are they guaranteed to be error-free, and I do not make any claims about the reliability or accuracy of these set of notes. They are intended primarily for personal use and for those who may benefit from an alternative presentation of the material.

Please use these notes at your own discretion.

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1 Linear Differential Equations with Variable Coefficients

2 Series Solutions and Special Functions

3 Sturm-Liouville Boundary Value Problems

4 Linear Systems of First Order Differential Equations

5 Laplace Transform Methods

6 Nonlinear Existence-Uniqueness Theory

7 Stability and Nonlinear Systems

7.1 Autonomous Systems

7.2 Phase Portraits of Linear Systems in $\mathbb{R}^2$

Let us consider the initial value problem

$$\mathbf{h}' = \mathbf{A}\mathbf{h}, \quad \mathbf{h}(0) = \mathbf{h}_0. \quad (1)$$

where $\mathbf{A}$ is some constant coefficient 2×2 matrix. In general, solving this IVP by the methods we know involves finding the eigenvalues of $\mathbf{A}$ and its corresponding eigenvectors. From that, we can construct the matrix $\mathbf{P}$ such that the Jordan form of $\mathbf{A}$ is $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$. Then, we find the solution $\mathbf{h}(t) = e^{\mathbf{A}t}\mathbf{h}_0 = \mathbf{P}e^{\mathbf{B}t}\mathbf{P}^{-1}\mathbf{h}_0$.

This is, however, quite annoying, and there exists an alternate method to do this. In fact, if we construct $\mathbf{P}$, and we let $\mathbf{z} = \mathbf{P}^{-1}\mathbf{h}$, then we know that the system $\mathbf{z}' = \mathbf{B}\mathbf{z}$ is satisfied. It is possible, then, to transform the solution $\mathbf{z}$ by the matrix $\mathbf{P}$ to construct the solution in $\mathbf{h}$.

Definition 7.2.1. Let $\mathbf{P}$ be invertible. If the linear transformation, $\mathbf{h} = \mathbf{P}\mathbf{z}$, maps the system $\mathbf{z}' = \mathbf{g}(\mathbf{z})$ into $\mathbf{h}' = \mathbf{f}(\mathbf{h})$, then the two systems and their corresponding phase portraits are *linearly equivalent*.

To study the application of this definition, we will reconsider the IVP we had before. In particular, we will let $\mathbf{A}$ be the 2×2 matrix with a Jordan form $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$. We now know that the phase portrait for the linear system $\mathbf{z}' = \mathbf{A}\mathbf{z}$ is equivalent to that of $\mathbf{h}' = \mathbf{B}\mathbf{h}$, so we can study all possible phase portraits of this system by considering the cases where $\mathbf{A}$ is in its Jordan form. The solution to the IVP is

$$\mathbf{h}(t) = e^{\mathbf{B}t}\mathbf{h}_0$$

and note that there are three possible forms of $\mathbf{B}$ depending on the eigenvalues of $\mathbf{A}$.

1. $\mathbf{B}$ has real eigenvalues, λ_1 and λ_2 , with linearly independent eigenvectors. That is,

$$\mathbf{B} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \implies e^{\mathbf{B}t} = \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix}.$$

2. $\mathbf{B}$ has a single, repeated eigenvalue, λ , with geometric multiplicity of 1. That is,

$$\mathbf{B} = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} = \lambda\mathbf{I} + \mathbf{N} \implies e^{\mathbf{B}t} = \begin{bmatrix} e^{\lambda t} & te^{\lambda t} \\ 0 & e^{\lambda t} \end{bmatrix}.$$

3. $\mathbf{B}$ has a pair of complex conjugate eigenvalues: $a \pm ib$, for $a, b \in \mathbb{R}$ and $b \neq 0$. That is,

$$\mathbf{B} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \implies e^{\mathbf{B}t} = e^{at} \begin{bmatrix} \cos(bt) & -\sin(bt) \\ \sin(bt) & \cos(bt) \end{bmatrix}.$$

Let us now classify the stability the equilibrium at the origin of each different phase portrait within $\mathbb{R}^2$. If $\lambda \neq 0$, we have the following:

Case I. $\mathbf{B}$ has real eigenvalues, both non-zero such that $\lambda_2 < 0 < \lambda_1$. We say that the equilibrium point is a *saddle point*. In this case, we have an unstable origin.

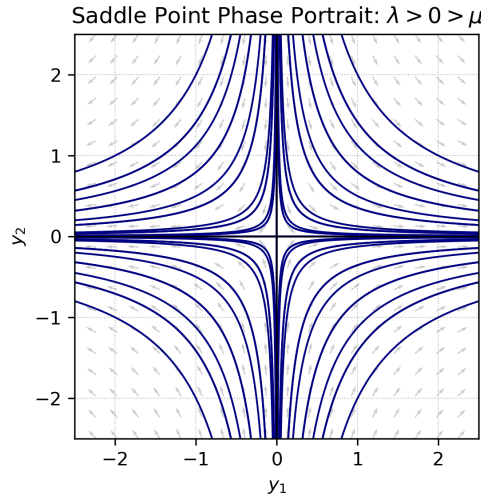


Figure 1: Phase portrait where the origin is a saddle point.

Case IIa. $\mathbf{B}$ has real eigenvalues λ_1 and λ_2 under the condition that $\lambda_1 \leq \lambda_2 < 0$, or $\mathbf{B}$ has a single, repeated eigenvalue, λ , with a geometric multiplicity of 1 under the condition $\lambda < 0$. In this case, we say that the equilibrium point is a *stable node*; and, furthermore, we can see that the solutions tend closer to the equilibrium point for all $t \geq 0$ and satisfy $\lim_{t \rightarrow \infty} \mathbf{h}(t) = \mathbf{0}$. It follows that the origin is asymptotically stable.

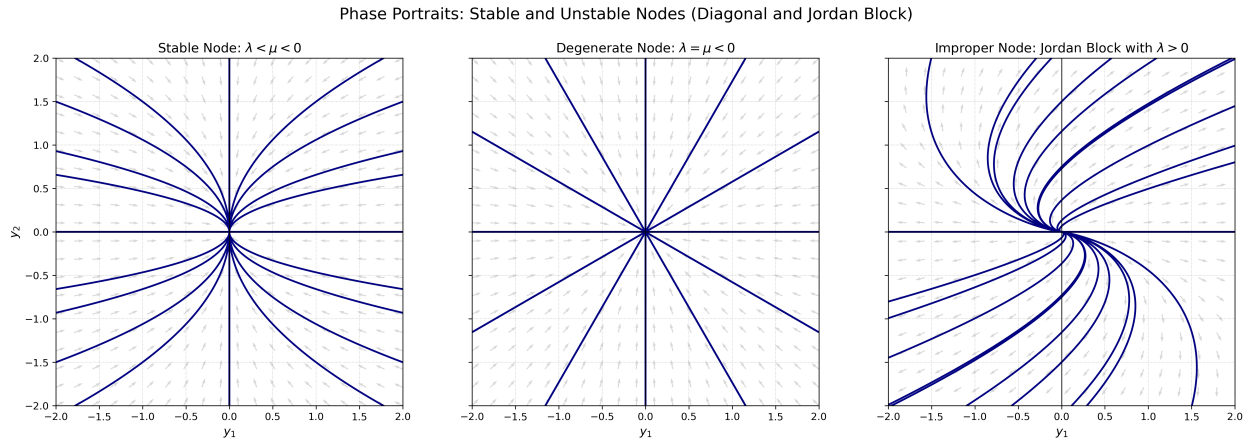


Figure 2: Three phase portraits of a stable node origin. The form of the phase portrait is determined by $\mathbf{B}$.

Case IIb. This case is similar to the case in IIa. However, the condition on λ_1 and λ_2 is such that $0 \leq \lambda_2 \leq \lambda_1$, and the condition on the single eigenvalue λ is such that $\lambda > 0$. Here, the equilibrium point is an *unstable node* and the directional arrows are reversed.

Case IIIa. $\mathbf{B}$ has a pair of complex conjugate eigenvalues, where in particular $a < 0$. In this case, we say that the equilibrium point is a *stable focus*, and the solutions tend to the equilibrium for all $t \geq 0$ and satisfy $\lim_{t \rightarrow \infty} \mathbf{h}(t) = \mathbf{0}$. It also follows that the origin is asymptotically stable.

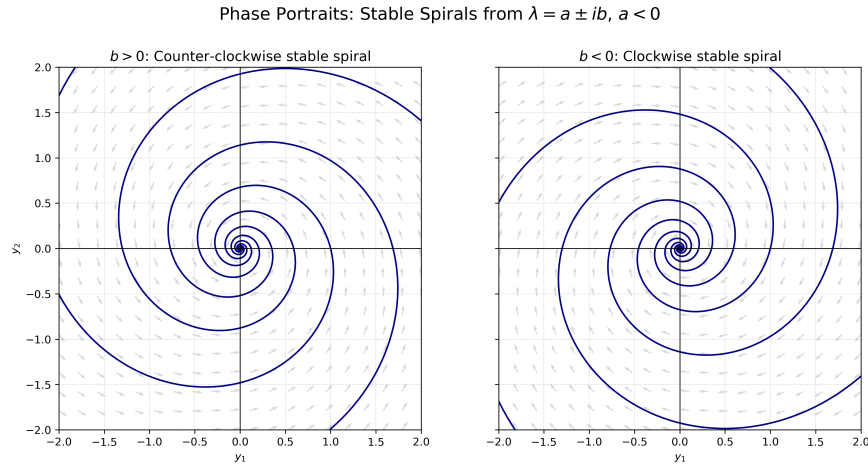


Figure 3: Two phase portraits of a stable focus corresponding to the case $a < 0$ and $b < 0$ or $b > 0$.

Case IIIb. This case is similar to the case in IIIa. However, the condition on a is such that $a > 0$. Here, the equilibrium point is an *unstable focus* and the directional arrows are reversed.

Case IV. $\mathbf{B}$ has a pair of complex conjugate eigenvalues where $a = 0$. In this case, we say that the origin is a *centre*. The equilibrium point is unique in the sense that any initial condition remains bounded, but solutions do not tend to $\mathbf{0}$ as $t \rightarrow \infty$. That is, the origin is stable but not asymptotically stable.

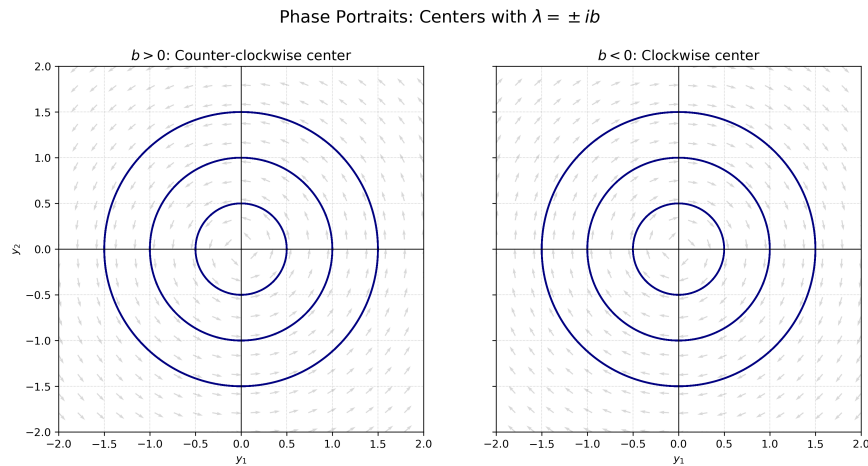


Figure 4: Two phase portraits of a centre equilibrium point corresponding to the cases $b < 0$ and $b > 0$.

You will notice that these cases can only be discussed after determining the Jordan form of $\mathbf{A}$. However, this is rather tedious and so we can extend our classifications to any system that satisfy

a coefficient matrix in Jordan form, so long as that system exists in $\mathbb{R}^2$. To do this, we will need some definitions.

Definition 7.2.2. The origin of the system $\mathbf{h}' = \mathbf{A}\mathbf{h}$ is said to be either a saddle, node focus, or centre if the phase portrait is linearly equivalent to that of Case(s) I, II, III, and IV.

Definition 7.2.3. The origin of the system $\mathbf{h}' = \mathbf{A}\mathbf{h}$ is said to be degenerate if $\det(\mathbf{A}) = 0$. That is, if $\mathbf{A}$ has at least one zero eigenvalue.

We can now determine how to classify the equilibrium point at the origin of our systems in $\mathbb{R}^2$. To do this, we need only to look at the determinant and the trace¹ of $\mathbf{A}$.

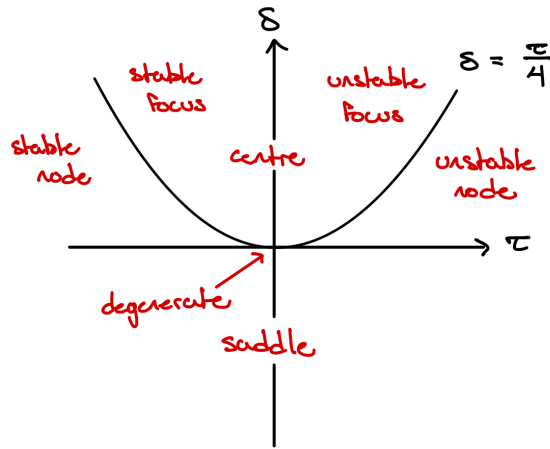


Figure 5: Classifications of the equilibrium point at the origin for the linear system $\mathbf{h}' = \mathbf{A}\mathbf{h}$ in $\mathbb{R}^2$. Here, $\tau = \text{Trace}(\mathbf{A})$ and $\delta = \det(\mathbf{A})$.

Theorem 7.2.1 (Classification of Linear Systems in $\mathbb{R}^2$). Let $\mathbf{A}$ be a 2×2 matrix for the linear system $\mathbf{h}' = \mathbf{A}\mathbf{h}$ and define $\tau = \text{Trace}(\mathbf{A})$ and $\delta = \det(\mathbf{A})$. Then,

- (a) If $\delta < 0$, the origin of the system is a saddle.
- (b) If $\delta > 0$ and $\tau^2 - 4\delta \geq 0$, the origin of the system is a node. Further, if $\tau < 0$, then the origin is a stable node, and if $\tau > 0$, the origin is an unstable node.
- (c) If $\delta > 0$, $\tau^2 - 4\delta < 0$ and $\tau \neq 0$, the origin of the system is a focus. Further, if $\tau < 0$, then the origin is a stable focus, and if $\tau > 0$, the origin is an unstable focus.
- (d) If $\delta > 0$ and $\tau = 0$, the origin of the system is a centre.
- (e) If $\delta = 0$, the origin of the system is degenerate.

Proof. The characteristic equation of $\mathbf{A}$ is

$$\lambda^2 - (a + d)\lambda + (ad - bc) = \lambda^2 - \tau\lambda + \delta = 0.$$

¹The trace of a matrix, $\text{Trace}(\mathbf{A})$, is the sum of the eigenvalues of the matrix.

Hence, the eigenvalues are

$$\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\delta}}{2}.$$

The results follow by considering each case of the eigenvalues given by the characteristic equation of $\mathbf{A}$. $\square$

7.3 Stability of Linear Systems

It is possible to generalize our linear system $\mathbf{h}' = \mathbf{A}\mathbf{h}$ to n dimensions by considering only the eigenvalues of $\mathbf{A}$.

Theorem 7.3.1. Let $\mathbf{A}$ be an $n \times n$ matrix. The equilibrium point, $\mathbf{h} = \mathbf{h}^* = \mathbf{0}$ of the linear system $\mathbf{h}' = \mathbf{A}\mathbf{h}$ is:

1. *asymptotically stable* if all the eigenvalues of $\mathbf{A}$ have a negative real part.
2. *stable*, but not asymptotically stable, if all the eigenvalues of $\mathbf{A}$ have a non-positive real part and any eigenvalues with a zero real part have the geometric multiplicity equal to the algebraic multiplicity.
3. *unstable* if at least one eigenvalue of $\mathbf{A}$ has a positive real part, or at least one eigenvalue with a zero real part has a geometric multiplicity less than the algebraic multiplicity.

7.4 Stability in Nonlinear Systems

Let us consider a nonlinear system of the form

$$\mathbf{h}' = \mathbf{f}(\mathbf{h}) \tag{2}$$

and suppose we are interested in the case of the system being in $\mathbb{R}^2$. Thus, we have

$$\mathbf{h} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} f(x, y) \\ g(x, y) \end{bmatrix}.$$

We can rewrite our system in its component form, such that

$$\begin{aligned} \frac{dx}{dt} &= f(x, y) \\ \frac{dy}{dt} &= g(x, y) \end{aligned}$$

where f and g are nonlinear functions of the variables x and y . The equilibrium solutions are determined by the null solution of the components, so

$$\begin{aligned} f(x^*, y^*) &= 0 \\ g(x^*, y^*) &= 0 \end{aligned}$$

for some constants x^* and y^* . We now consider a solution that is near the equilibrium solution, such that

$$x(t) = x^* + u(t), \quad y(t) = y^* + v(t)$$

for some small functions $u(t)$ and $v(t)$. We are interested in how small perturbations in this solution evolve in time, so we can find equations for $u(t)$ and $v(t)$ in particular. To do this, let us substitute the equations into the nonlinear functions and invoke the Taylor remainder theorem, such that

$$\begin{aligned} f(x, y) &= f[x^* + u(t), y^* + v(t)] \\ &= \cancel{f(x^*, y^*)} + u(t) \left. \frac{\partial f}{\partial x} \right|_{(x^*, y^*)} + v(t) \left. \frac{\partial f}{\partial y} \right|_{(x^*, y^*)} + r(u, v) \\ &= u(t) \left. \frac{\partial f}{\partial x} \right|_{(x^*, y^*)} + v(t) \left. \frac{\partial f}{\partial y} \right|_{(x^*, y^*)} + r(u, v) \end{aligned}$$

where the function $f(x^*, y^*) = 0$ by our imposed condition and our remainder, $r(u, v)$ satisfies the limit

$$\lim_{(u(t), v(t)) \rightarrow (0, 0)} \frac{r[u(t), v(t)]}{\sqrt{u^2(t) + v^2(t)}} = 0.$$

Assuming that both parameters $|u(t)|, |v(t)| \ll 1$, the remainder term also is much smaller than the linear terms, and so we can approximate the first two non-zero terms in our expansion to be

$$\begin{aligned} g(x, y) &\approx u(t) \left. \frac{\partial f}{\partial u} \right|_{(x^*, y^*)} + v(t) \left. \frac{\partial f}{\partial v} \right|_{(x^*, y^*)} \\ g(x, y) &\approx u(t) \left. \frac{\partial g}{\partial u} \right|_{(x^*, y^*)} + v(t) \left. \frac{\partial g}{\partial v} \right|_{(x^*, y^*)}. \end{aligned}$$

Notice that we have switched the derivative argument from $(x, y) \rightarrow (u, v)$, since the approximation can assume that $(x, y) \approx (u, v)$. On the other hand, the left side of the equations in terms of our perturbations become

$$\begin{aligned} \frac{dx}{dt} &= \frac{d(x^* + u)}{dt} = \frac{du}{dt} \\ \frac{dy}{dt} &= \frac{d(y^* + v)}{dt} = \frac{dv}{dt} \end{aligned}$$

by similar arguments as before. In this way, we arrive at the approximation system of linearized differential equations,

$$\begin{aligned} \frac{du}{dt} &= u \left. \frac{\partial f}{\partial u} \right|_{(x^*, y^*)} + v \left. \frac{\partial f}{\partial v} \right|_{(x^*, y^*)} \\ \frac{dv}{dt} &= u \left. \frac{\partial g}{\partial u} \right|_{(x^*, y^*)} + v \left. \frac{\partial g}{\partial v} \right|_{(x^*, y^*)}. \end{aligned}$$

In vector form, this is rewritten as

$$\frac{d}{dt} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \partial f / \partial u & \partial f / \partial v \\ \partial g / \partial u & \partial g / \partial v \end{bmatrix}_{(x^*, y^*)} \begin{bmatrix} u \\ v \end{bmatrix}.$$

Notice that this is a constant coefficient linear system, and the coefficient matrix is the *Jacobian matrix* of $\mathbf{f} = (f, g)^T$. We thus expect that the behaviour of solutions near the equilibrium point, (x^*, y^*) , to behave like this system.

We can, in fact, formalize this for a general nonlinear system in $\mathbb{R}^n$, such that $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$.

Definition 7.4.1. The linearization of the nonlinear system $\mathbf{h}' = \mathbf{f}(\mathbf{h})$ about the equilibrium point $\mathbf{h}^*$, where $\mathbf{f}(\mathbf{h}^*) = 0$, is the constant coefficient linear system

$$\mathbf{h}' = \mathbf{J}_f(\mathbf{h}^*). \quad (3)$$

With this new definition, we can determine the behaviour of our solutions of the nonlinear system near the equilibrium point. Focusing primarily on determining stability, we have the following theorem.

Theorem 7.4.1 (Stability of Equilibria by Linearization). Let $\mathbf{f} \in C^1(E)$ where $(\mathbf{h}^* \in E) \subset \mathbb{R}^n$. Then,

- (a) if every eigenvalue of $\mathbf{J}_f(\mathbf{h}^*)$ has a negative real part, then $\mathbf{h}^*$ is asymptotically stable.
- (b) if at least one eigenvalue of $\mathbf{J}_f(\mathbf{h}^*)$ has a positive real part, then $\mathbf{h}^*$ is unstable.

The proof is in AMATH 451, sadly. It should be noted that the only condition on Theorem 7.4.1 is that $\mathbf{f}$ is C^1 . Another important point is that if $\mathbf{J}_f(\mathbf{h}^*)$ has some eigenvalues with a negative real part and some with a zero real part, then the theorem is inconclusive. Why? The linear terms no longer dominate the nonlinear terms even near the origin, so linearization fails to predict behaviour.

A general strategy to determine the stability of equilibrium solutions of our nonlinear system can be broken down into four parts.

1. Compute the equilibrium solutions by solving the case $\mathbf{f}(\mathbf{h}^*) = \mathbf{0}$ for $\mathbf{h}^*$.
2. Compute the Jacobian matrix $\mathbf{J}_f(\mathbf{h})$ and evaluate it at each equilibrium point.
3. Compute the eigenvalues of the Jacobian at $\mathbf{h}^*$ (may be multiple points).
4. Apply Theorem 7.4.1 to determine stability.

In $\mathbb{R}^2$, we can also state more specific results on how the properties of our phase portraits carry over to nonlinear systems close to equilibria.

Theorem 7.4.2 (Nonlinear Saddle Points). Let $\mathbf{f} \in C^1(E)$ where $(\mathbf{h}^* \in E) \subset \mathbb{R}^2$. If $\mathbf{J}_f(\mathbf{h}^*)$ has one positive eigenvalue and one negative eigenvalue, then $\mathbf{h}^*$ is a *saddle point* of the nonlinear system $\mathbf{h}' = \mathbf{f}(\mathbf{h})$. Further, the directions of the solutions moving to or away from $\mathbf{h}^*$ is given by the eigenvectors of $\mathbf{J}_f(\mathbf{h}^*)$.

Theorem 7.4.3 (Nonlinear Nodes and Foci). Suppose there is nonlinear system is given by $\mathbf{h}' = \mathbf{f}(\mathbf{h})$. Let $\mathbf{f} \in C^2(E)$ where $\mathbf{h}^* \in E) \subset \mathbb{R}^2$.

1. If $\mathbf{J}_f(\mathbf{h}^*)$ has two negative eigenvalues, then $\mathbf{h}^*$ is a stable node of the nonlinear system.
2. If $\mathbf{J}_f(\mathbf{h}^*)$ has two positive eigenvalues, then $\mathbf{h}^*$ is a unstable node of the nonlinear system.
3. If $\mathbf{J}_f(\mathbf{h}^*)$ has two complex conjugate eigenvalues with a negative real part, then $\mathbf{h}^*$ is a stable focus of the nonlinear system.
4. If $\mathbf{J}_f(\mathbf{h}^*)$ has two complex conjugate eigenvalues with a positive real part, then $\mathbf{h}^*$ is a stable focus of the nonlinear system.

7.4.1 The Nonlinear Pendulum

Let us consider the model of the damped, nonlinear pendulum. The governing DE is

$$\frac{d^2\theta}{dt^2} + c\frac{d\theta}{dt} + \omega_0^2 \sin(\theta) = 0$$

and we can write this as a system by defining $x = \theta$ and $y = d\theta/dt$,

$$\begin{aligned}\frac{dx}{dt} &= y \\ \frac{dy}{dt} &= -\omega_0^2 \sin(x) - cy.\end{aligned}$$

In general, we cannot find analytical solutions in terms of elementary functions. We can, however, determine the equilibrium solutions. In this case, we set the time derivatives to zero and find that

$$\begin{aligned}y &= 0 \\ -\omega_0^2 \sin(x) - cy &= 0.\end{aligned}$$

It follows that the possible equilibrium solutions are $y = d\theta/dt = 0$ and $x = \theta = n\pi$. However, we will restrict our domain to $\theta \in [0, 2\pi]$, and so we have the solutions

$$\theta = 0, \quad \theta = \pm\pi.$$

Intuitively, we know these solutions are very different. The first solution corresponds to zero movement at the point $(x, y) = (0, 0)$. That is, the pendulum hangs straight down. The other solution corresponds to the pendulum being inverted. Furthermore, we may expect that the second solution is unstable, since the rod will always tend to fall. These cases hold for both the undamped ($c = 0$) case, and the damped case ($c \neq 0$).

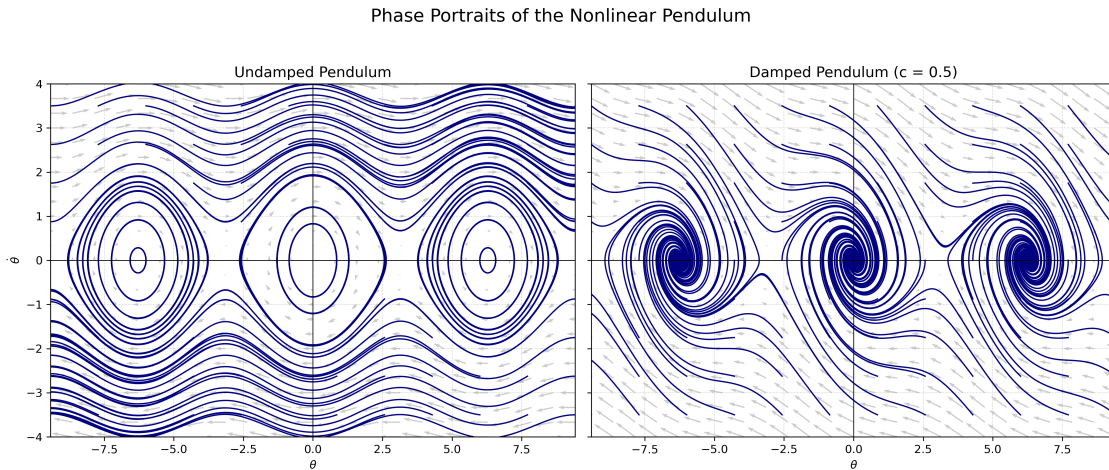


Figure 6: Phase portrait of the nonlinear pendulum in both the damped and undamped cases.

Let us now investigate the equilibrium points of the system in the undamped case. In particular, the solutions are given by the points

$$(x^*, y^*) = (0, 0) \quad \text{and} \quad (x^*, y^*) = (\pm\pi, 0).$$

We can compute the Jacobian matrix of the system if we recognize that

$$\frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} f(x, y) \\ g(x, y) \end{bmatrix} = \begin{bmatrix} y \\ -\omega_0^2 \sin(x) \end{bmatrix}.$$

Naturally, we arrive at the matrix

$$\mathbf{J}_f(\mathbf{h}) = \begin{bmatrix} \partial f / \partial x & \partial f / \partial y \\ \partial g / \partial x & \partial g / \partial y \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 \cos(x) & 0 \end{bmatrix}$$

where, again, we stress that $(x, y) \approx (u, v)$. Computing the Jacobian at the equilibrium solutions, we arrive at

$$\begin{aligned} \mathbf{J}_f(\mathbf{h}^*) &= \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & 0 \end{bmatrix}, & (x^*, y^*) &= (0, 0) \\ \mathbf{J}_f(\mathbf{h}^*) &= \begin{bmatrix} 0 & 1 \\ \omega_0^2 & 0 \end{bmatrix}, & (x^*, y^*) &= (\pm\pi, 0). \end{aligned}$$

We now must compute the eigenvalues of our matrix. For the case that $(x^*, y^*) = (0, 0)$, we simply have

$$\begin{vmatrix} \lambda & -1 \\ \omega_0^2 & \lambda \end{vmatrix} = \lambda^2 + \omega_0^2 \implies \lambda = \pm i\omega_0$$

where we assume that $\omega_0 > 0$. Notice that these eigenvalues are complex conjugate with a zero real part. Thus, Theorem 7.4.1 is inconclusive and we cannot determine the stability of the equilibrium point at $(0, 0)$.

Looking at the case $(x^*, y^*) = (\pm\pi, 0)$, we find the determinant and eigenvectors to be

$$\begin{vmatrix} \lambda & -1 \\ -\omega_0^2 & \lambda \end{vmatrix} = \lambda^2 - \omega_0^2 \implies \lambda = \pm\omega_0, \quad \text{and} \quad \mathbf{v}_{1,2} = \begin{bmatrix} \pm 1 \\ \omega_0 \end{bmatrix}.$$

Here, the eigenvalues are both real with a positive and negative part. Thus, by Theorem 7.4.1, we can conclude that these equilibrium points are unstable. By Theorem 7.4.2, we can further determine that this equilibrium point is a saddle point. The eigenvectors of this case differ by only the first component ± 1 ; we find that the eigenvalue $\lambda_1 = \omega_0$ corresponds to $\mathbf{v}_1 = [1, \omega_0]^T$, suggesting that the direction of the trajectory travels to the upper right.

On the other hand, the eigenvalue $\lambda_2 = -\omega_0$ corresponds to $\mathbf{v}_2 = [-1, \omega_0]^T$, which suggests the trajectory travels to the bottom left. Of course, it follows that the other trajectories approach conversely; from the top left and from the bottom right.

Comparing the predicted behaviour to the numerically generated phase portrait in Fig. 7, it can be seen that the saddle point behaves as expected. The origin point, however, appears to be a stable centre but not asymptotically stable; this could not have been predicted. We can, however, make the connection that this centre behaviour of the origin arises from the conservative property of the nonlinear pendulum. That is, it conserves energy.

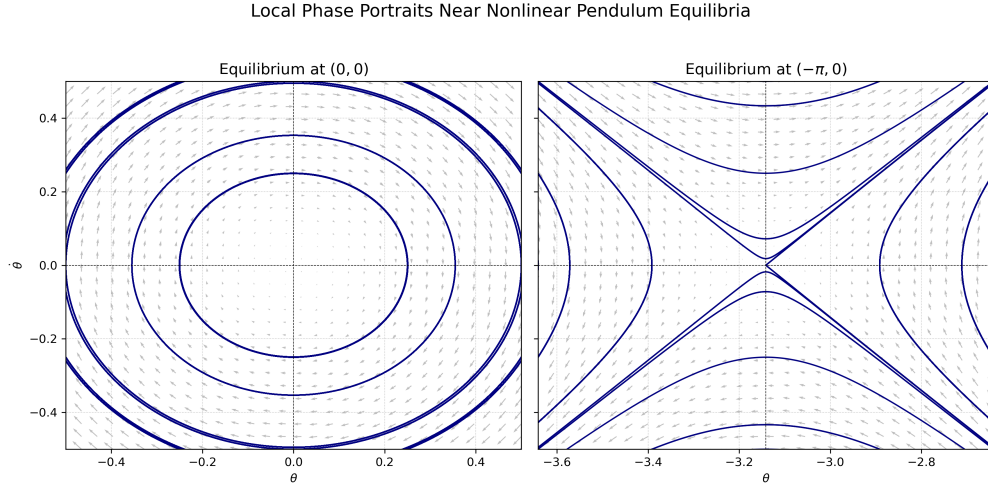


Figure 7: Phase portraits around the two equilibrium solutions $(0,0)$ and $(-\pi,0)$ for the nonlinear undamped pendulum.

7.4.2 The Simple Competing Species Model

Let us analyze a simple model for two species compete for the same food supply. We can define $x(t)$ and $y(t)$ to be the populations of each species, and the interaction of them is described by

$$\begin{aligned}\frac{dx}{dt} &= x(a_1 - b_1x - c_1y) \\ \frac{dy}{dt} &= y(a_2 - b_2y - c_2x)\end{aligned}$$

for some positive constants a_1, a_2, b_1, b_2, c_1 and c_2 which determine the interactions of the species. Specifically, breaking down the model into their essential components, we have:

1. that a_1 represents the growth rate of the species, $x(1)$.
2. that b_1 represents the mortality rate of $x(t)$.
3. that c_1 represents the rate at which species $x(t)$ does due to competition with $y(t)$.
4. that a_2 represents the growth rate of species $y(t)$.
5. that b_2 represents the mortality rate of $y(t)$.
6. that c_2 represents the rate at which species $y(t)$ does due to competition with $x(t)$.

In the case where neither species interacts, we have that $c_1 = c_2 = 0$, which simplifies into the Logistic growth model. We can determine the equilibrium solutions by setting the derivatives to zero, such that

$$\begin{aligned}x(a_1 - b_1x - c_1y) &= 0 \\ y(a_2 - b_2y - c_2x) &= 0.\end{aligned}$$

There are four possible equilibrium solutions, which are

$$(0, 0), \quad \left(0, \frac{a_2}{b_2}\right), \quad \left(\frac{a_1}{b_1}, 0\right), \quad (x^*, y^*)$$

and we can solve for the equilibrium points in the last ordered pair,

$$x^* = \frac{a_1 b_2 - a_2 c_1}{b_1 b_2 - c_1 c_2}$$

$$y^* = \frac{b_1 a_2 - a_1 c_2}{b_1 b_2 - c_1 c_2}.$$

Notice that each equilibrium pair describes a different biological state in the system. The first ordered pair, $(0, 0)$, corresponds to the case where neither species is present. The second pair, $(0, a_2/b_2)$, corresponds to the case where the species $y(t)$ is present but $x(t)$ is not. Conversely, $(a_1/b_1, 0)$ corresponds to the case where $x(t)$ is present and $y(t)$ is not, and the fourth pair (x^*, y^*) corresponds to the case where both species are present.

We can immediately determine the qualitative nature of the phase portrait by analyzing the signs of dx/dt and dy/dt by looking at the nullclines. Of course, the solution is only physically meaningful when $(x \wedge y) \geq 0$, so we will look at the first quadrant of our portrait. From our derivatives, we have the equations

$$0 = x(a_1 - b_1 x - c_1 y) \implies y = \frac{a_1}{c_1} - \frac{b_1}{c_1} x, \quad \text{or} \quad x = 0$$

$$0 = y(a_2 - b_2 y - c_2 x) \implies y = \frac{a_2}{b_2} - \frac{c_2}{b_2} x, \quad \text{or} \quad y = 0.$$

We thus have four nullclines. Two of them are trivial, which are the cases $x = y = 0$. On the other hand, the case

$$y = \frac{a_1}{c_1} - \frac{b_1}{c_1} x$$

starts at the point $(0, a_1/c_1)$ and finishes at $(a_1/b_1, 0)$. Notice, if $x > 0$, then we see that below the line, $dx/dt > 0$, while above $dx/dt < 0$. We thus expect that trajectories below the line will move to the right, while trajectories above the line will move to the left.

On the other hand, the second equation,

$$y = \frac{a_2}{b_2} - \frac{c_2}{b_2} x$$

starts at the point $(0, a_2/b_2)$ and finishes at $(a_2/c_2, 0)$. When $y > 0$, we see that in the case below the line, $dy/dt > 0$, while above the line $dy/dt < 0$. That is, trajectories below the line move upward while trajectories above the line move downward.

There are four different unique cases for the two linear nullclines. Discussing them here, we have: (1) The nullclines do not intersect and we have that the first nullcline is under the second. (2) The nullclines do not intersect and the first nullcline is over the second. (3) The nullclines intersect, and at $x = 0$ the first nullcline is over the second. (4) The nullclines intersect, and at $x = 0$ the first nullcline is under the second.

Let us now look at the stability of our solutions. To do this, we can write the Jacobian, which is

$$\mathbf{J}_f(\mathbf{h}^*) = \begin{bmatrix} \partial f / \partial x & \partial f / \partial y \\ \partial g / \partial x & \partial g / \partial y \end{bmatrix}_{(x^*, y^*)} = \begin{bmatrix} a_1 - 2b_1 x - c_1 y & -c_1 x \\ -c_2 y & a_2 - 2b_2 y - c_2 x \end{bmatrix}_{(x^*, y^*)}.$$

Now, we must look at each individual equilibrium point.

Case (0,0). The Jacobian matrix becomes

$$\mathbf{J}_f(\mathbf{h}^*) = \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix}$$

which is a diagonal matrix in terms of its eigenvalues. Notice, both eigenvalues are positive and the eigenvectors are in terms of the standard basis, so Theorem 7.4.1 shows that the origin is unstable. We can further apply Theorem 7.4.3, which shows that the origin is also an unstable node.

Case (0, a_2/b_2). Here, the Jacobian matrix is

$$\mathbf{J}_f(\mathbf{h}^*) = \begin{bmatrix} a_1 - c_1 a_2/b_2 & 0 \\ -c_2 a_2/b_2 & -a_2 \end{bmatrix}$$

to which the eigenvalues are $\lambda_1 = a_1 - c_1 a_2/b_2$ and $\lambda_2 = -a_2$. Notice, assuming that the parameters are non-zero, the eigenvalues are both real and that $\lambda_2 < 0$. There are now two cases of possible stabilities. By Theorems 7.4.1-7.4.3, if $\lambda_1 > 0$, the equilibrium point is a saddle, while if $\lambda_1 < 0$ the equilibrium point is a stable node.

Case (a_1/b_1 , 0). Here, the Jacobian matrix is

$$\mathbf{J}_f(\mathbf{h}^*) = \begin{bmatrix} -a_1 & -a_1 c_1/b_1 \\ 0 & a_2 - c_2 a_1/b_1 \end{bmatrix}$$

to which our eigenvalues are $\lambda_1 = -a_1$ and $\lambda_2 = a_2 - c_2 a_1/b_1$. Similar to the previous case, these eigenvalues are both real, and $\lambda_1 < 0$ assuming that the parameters are non-zero. By Theorems 7.4.1-7.4.3, we have that $\lambda_2 > 0$ is a saddle point, and $\lambda_2 < 0$ is a stable node.

We will not discuss the last case, as the Jacobian is rather difficult to analyze. However, we will look at our analyses numerically. Choosing the parameters $a_1 = b_1 = b_2 = 1$, $a_2 = 3/4$ and $c_2 = 1/2$, we have the equilibrium solutions

$$(x, y) = (0, 0), \quad (x, y) = \left(0, \frac{3}{4}\right), \quad (x, y) = (1, 0), \quad (x, y) = \left(\frac{1}{2}, \frac{1}{2}\right).$$

This particular choice of constants corresponds to case (3), where the nullclines intersect and at $x = 0$ the first is over the second. Notice that the point (0,0) is an unstable node as predicted, since the directional arrows tend to move away from the equilibrium, and the other points behave as predicted.

On the other hand, choosing the constants $a_1 = b_1 = c_1 = 1$, $a_2 = 1/2$, $b_2 = 1/4$, and $c_2 = 3/4$ gives the equilibrium solutions

$$(x, y) = (0, 0), \quad (x, y) = (0, 2), \quad (x, y) = (1, 0), \quad (x, y) = \left(\frac{1}{2}, \frac{1}{2}\right).$$

Furthermore, the second and third pairs are stable nodes while the fourth pair is an unstable saddle point.

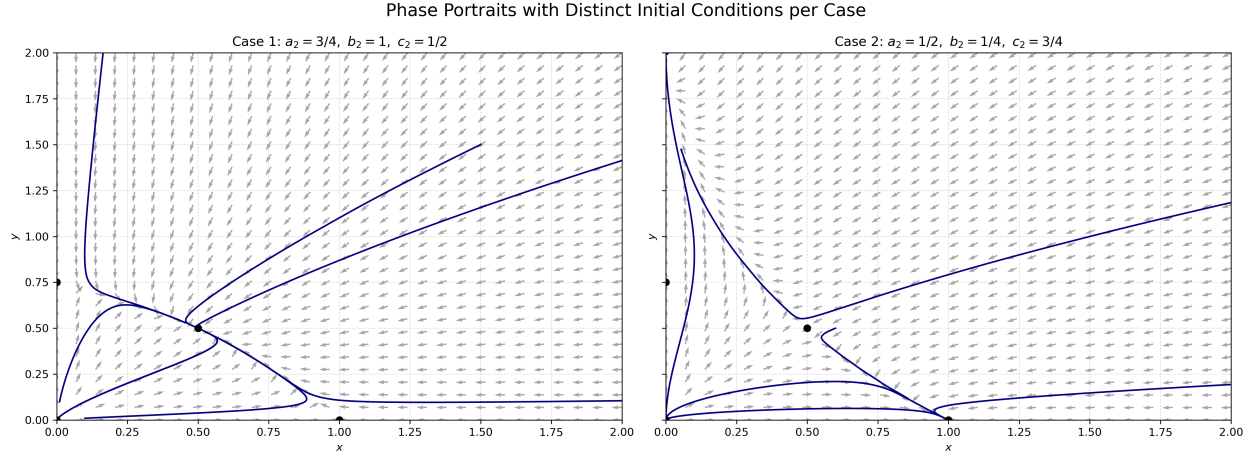


Figure 8: Two phase portraits of the simple competing species model for differing parameters.

This system, in particular, as an example of a **bistable system**. That is, there are two stable equilibrium points; all solutions will tend to one or the two stable points depending on the initial conditions of the trajectories. The region within which initial conditions lead to a particular equilibrium point is called the **basin of attraction**. There is a specific curve which separates the basins which we call the **separatrix**, corresponding to the initial conditions for which the solutions tend to the saddle point.

7.5 Periodic Solutions

A unique class of solutions which occur in several models are periodic solutions. That is, solutions which satisfy the condition $\mathbf{h}(t + T) = \mathbf{h}(t)$ for some $T > 0$ and $t \in \mathbb{R}$. The constant, T , is the *period* of the solution, and a periodic solution itself appears as a closed curve. We call these *cycles* or *periodic orbits*.

7.5.1 Isolated Circular Orbits

Let us consider a system of the form

$$\begin{aligned}\frac{dx}{dt} &= x - y - x(x^2 + y^2) \\ \frac{dy}{dt} &= x + y - y(x^2 + y^2).\end{aligned}$$

The equilibrium point of this system only occurs at the origin, $(x, y) = (0, 0)$, and so our Jacobian matrix is

$$\mathbf{J}_f(\mathbf{h}^*) = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad \lambda = 1 \pm i.$$

which, by Theorem 7.4.3, is an unstable focus. Furthermore, if we apply a polar coordinate transformation, we have that

$$\begin{aligned}r \frac{dr}{dt} &= x \frac{dx}{dt} + y \frac{dy}{dt} = (x^2 + y^2) - (x^2 + y^2)^2 = r^2 - r^4 \\ \frac{d\theta}{dt} &= \frac{x^2}{x^2 + y^2} \left[\frac{1}{x} \frac{dy}{dt} - \frac{y}{x^2} \frac{dx}{dt} \right] = \frac{1}{x^2 + r^2} \left[x \frac{dy}{dt} - y \frac{dx}{dt} \right]\end{aligned}$$

where we have taken the derivatives of

$$r^2 = x^2 + y^2, \quad \theta = \arctan\left(\frac{y}{x}\right).$$

Applying the transformations, we arrive at

$$\begin{aligned} \frac{dr}{dt} &= r(1 - r^2) \\ \frac{d\theta}{dt} &= 1. \end{aligned}$$

Notice we now have the nullclines in r governed by the equations

$$r = 0, \quad r = 1.$$

The case $r = 0$ corresponds to the origin, and the case $r = 1$ corresponds to a circle of radius 1. Notice also that the nullcline in θ does not exist. That is, is always the constant 1; we expect θ to thus increase linearly in t ,

$$\theta(t) = t + C$$

for some constant C . Let us discuss that case $r = 1$. We know, in fact, that since the derivatives are zero, the case $r = 1$ must be a constant periodic orbit of radius 1. However, this constant periodic orbit is the *only* periodic orbit. If we investigate the phase portrait in polar coordinates, we have that

$$\begin{aligned} \frac{dr}{dt} &> 0 \quad \text{if } r < 1 \\ \frac{dr}{dt} &< 0 \quad \text{if } r > 1. \end{aligned}$$

We thus expect that r increases if $r < 1$ and r decreases if $r > 1$. Furthermore, since $\theta(t)$ increases linearly with t , the phase space in the $r\theta$ -plane shows that the case $r > 1$ has counter-clockwise, decreasing radii trajectories, while the case $r < 1$ has counter-clockwise, increasing radii trajectories.

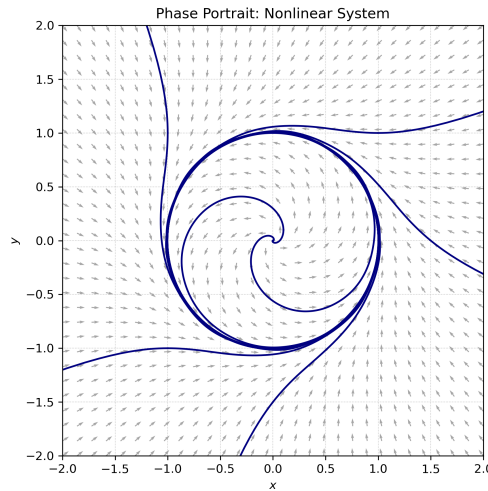


Figure 9: Phase portrait for the isolated circular orbit.

Notice, in fact, that the qualitative analysis of the phase portrait suggests that every trajectory tends to approach the periodic orbit as $t \rightarrow \infty$. In fact, we will define (heuristic) definitions of a periodic orbit's stability based on the phase portrait of our systems.

Definition 7.5.1. A periodic orbit is:

1. *stable* if every trajectory that starts close enough to it remains close for all $t \geq 0$.
2. *unstable* if it is not stable.
3. *asymptotically stable* if it is stable and every trajectory that starts close enough to it approaches it as $t \rightarrow \infty$.

In general, it is difficult to find periodic orbits analytically. However, we can formulate theorems for their existence in the $\mathbb{R}^2$ case.

Theorem 7.5.1 (Periodic Orbits Surrounding Equilibrium Points). Let $\mathbf{f} = [f(x, y), g(x, y)]^T \in C^1(E)$ where $E \subset \mathbb{R}^2$. A periodic orbit of the nonlinear system $\mathbf{h}' = \mathbf{f}(\mathbf{h})$ in E must enclose at least one equilibrium point. If there is only a single equilibrium point enclosed by a periodic orbit, then it cannot be a saddle point.

Theorem 7.5.2 (Dulac's Criteria). Let $\mathbf{f} = [f(x, y), g(x, y)]^T \in C^1(E)$ where $E \subset \mathbb{R}^2$ and E is simply connected. If there is a real valued function, $B \in C^1(E)$, such that $\nabla \cdot (B\mathbf{f})$ is not identically zero and does not change sign in E , then the nonlinear system $\mathbf{h}' = \mathbf{f}(\mathbf{h})$ has no closed orbit lying entirely in E .

7.6 Bifurcations

Much of our discussion on the stability of equilibrium points has hinged on the parameters in our models. For example, the simple competing species model resulted in different types of stability depending on the values of a_1, b_1, c_1, a_2, b_2 and c_2 . This is a more general result of a what we call a *bifurcation*. If a parameter in our model is allowed to vary, then at some point we have that the phase portraits become different in image; we say that the value of the parameter at the change is called the **bifurcation value**.

If we recall Theorem 7.4.2, we can see that the stability of our system changes depending on the real part of the eigenvalues corresponding to $\mathbf{J}_f(\mathbf{h}^*)$. In particular, the change occurs from the real part switching from positive to negative, and so we can infer the bifurcations are mostly associated with eigenvalues of a zero real part. We call these **non-hyperbolic** equilibrium points.

7.6.1 Saddle Node Bifurcation

Let us consider the system

$$\begin{aligned}\frac{dx}{dt} &= \mu + x^2 \\ \frac{dy}{dt} &= -y.\end{aligned}$$

The equilibrium points correspond to three different cases of μ . In particular,

$$\begin{aligned}\mu < 0 &\implies (-\sqrt{-\mu}, 0) = \mathbf{x}_{\pm} \\ \mu = 0 &\implies (0, 0) = \mathbf{0} \\ \mu > 0 &\implies \text{none.}\end{aligned}$$

If we have the parameter, μ , such that $\mu < 0$, the Jacobian matrix is

$$\mathbf{J}_f(\pm\sqrt{-\mu}, 0) = \begin{bmatrix} \pm 2\sqrt{-\mu} & 0 \\ 0 & -1 \end{bmatrix}$$

where x_+ is a saddle point and x_- is a stable node. In the case $\mu = 0$, our Jacobian becomes

$$\mathbf{J}_f(0, 0) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

with has an eigenvalue of zero and therefore $(0, 0)$ is non-hyperbolic. Thus, $\mu = 0$ corresponds to a bifurcation. Since x_+ is a saddle point and switches to a stable node, we say that the parameter $\mu = 0$ corresponds to a **saddle-node bifurcation**.

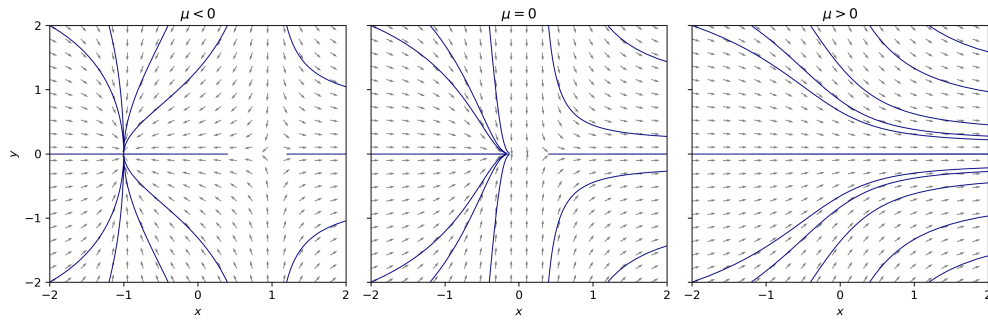


Figure 10: Phase portraits for the saddle node bifurcation in each case of μ .

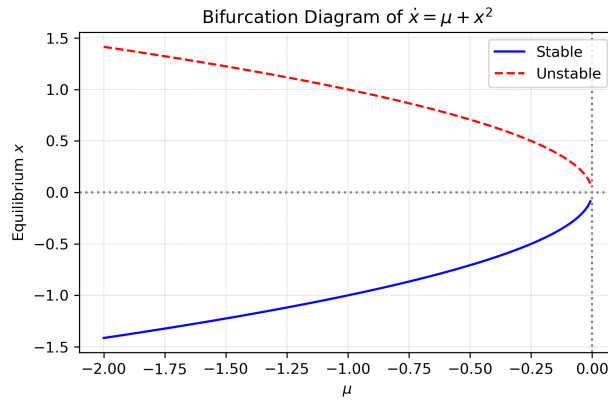


Figure 11: Bifurcation diagram for the saddle-node bifurcation. The dashed line corresponds to an unstable equilibrium point, while the solid line is an asymptotically stable equilibrium point.

8 Perturbation Methods

To begin discussing perturbation methods, let us begin with a motivating example. We will consider an IVP of the form

$$y'' + (1 + \epsilon)y = 0, \quad y(0) = 1, \quad y'(0) = 0$$

where we assume that $|\epsilon| < 1$. The solution of this equation is of the form

$$y(t; \epsilon) = \cos(\sqrt{1 + \epsilon}t).$$

Furthermore, since the solution is analytic at $t = 0$, we can express the solution in the form of a power series in t , such that

$$y(t; \epsilon) = \sum_{k=0}^{\infty} \frac{(-1)^k (1 + \epsilon)^k}{k!} t^{2k}.$$

Notice that this solution is of the form $y(t) = \sum_{k=0}^{\infty} a_k(\epsilon)t^k$, where the coefficients of the power series depend on the parameter, ϵ .

To find an approximation to the solution, we need only to truncate the series. For example, we may approximate the solution with only the first two terms, which turns out to be a Taylor polynomial about $t_0 = 0$ of degree 2,

$$y(t; \epsilon) \approx 1 - \frac{1 + \epsilon}{2} t^2 = T_2(t).$$

Taylor's theorem shows that the error in the approximation is

$$|y(t; \epsilon) - T_2(t)| \leq C(\epsilon)t^3$$

where $C(\epsilon)$ is some parameter which depends on ϵ . In order notation, we can equivalently express this as

$$y(t) = T_2(t) + \mathcal{O}(t^3).$$

Notice that this approximation is only good when t is close to zero. We can easily improve this approximation by adding more terms, but equivalently the error in the approximation grows very large once $|t| > 1$.

Perturbation theory attempts to circumvent this issue by expressing the solution as a power series in ϵ ,

$$y(t; \epsilon) = \sum_{k=0}^{\infty} \frac{1}{k!} \left. \frac{d^k y}{d\epsilon^k} \right|_{\epsilon=0} \epsilon^k.$$

If we proceed with the differentiation on the solution, the first two terms in the approximate solution are

$$T_1(\epsilon) = \cos(t) - \frac{t \sin(t)}{2} \epsilon$$

and we can apply Taylor's Theorem to show that

$$|y(t; \epsilon) - T_1(\epsilon)| \leq \bar{C}(t)\epsilon^2$$

for some parameter $\bar{C}(t)$ which is dependent on t . Notice the difference between this approximate solution and the previous approximate solution. The series in ϵ is *still* good even if ϵ is close enough to zero, *and* the solution improves as ϵ get smaller. The caveat, however, is that the bound on the error depends on t itself.

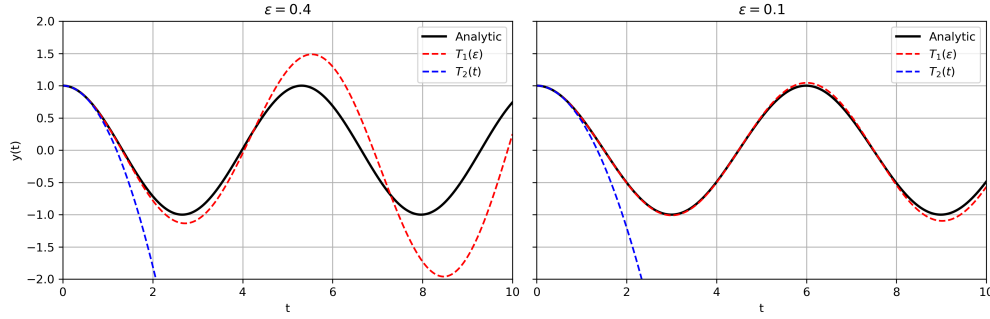


Figure 12: Comparison of the approximations $T_2(t)$ and $T_1(\epsilon)$ to the analytic solution $y(t; \epsilon) = \cos(\sqrt{1 + \epsilon}t)$ for varying ϵ .

We can extend this approach to arbitrary functions, which is the idea of perturbation theory. Let us assume that there is a solution in the form of a power series

$$y(t; \epsilon) = \sum_{k=0}^{\infty} y_k(t) \epsilon^k$$

we can find the coefficients $y_k(t)$ by substituting them into the differential equation. The main idea is that if the coefficients, $y_k(t)$, are well-behaved, then for a given value of ϵ we can find an approximation to the solution that is valid for a large range of t values.

8.1 Regular Perturbation Theory

8.1.1 Algebraic Equations

Let us consider the algebraic equation

$$x^2 + (4 + \epsilon)x + 3 - 2\epsilon = 0.$$

We can, by the quadratic formula, determine the two exact roots of the equations, which are

$$x_+(\epsilon) = \frac{-(4 + \epsilon) + \sqrt{(4 + \epsilon)^2 - 12 + 8\epsilon}}{2}$$

$$x_-(\epsilon) = \frac{-(4 + \epsilon) - \sqrt{(4 + \epsilon)^2 - 12 + 8\epsilon}}{2}.$$

To apply perturbation methods, we can approach this problem by using the basic idea: assuming that $|\epsilon| \ll 1$, we look for an approximate solution to the quadratic equation of the form

$$x(\epsilon) = x_0 + \epsilon x_1 + \epsilon^2 x_2 + \cdots + \epsilon^n x_n.$$

Substituting this solution in into the equation and gathering like powers of ϵ , we have

$$(x_0 + \epsilon x_1 + \epsilon^2 x_2 + \cdots)^2 + (4 + \epsilon)(x_0 + \epsilon x_1 + \epsilon^2 x_2 + \cdots) + 3 - 2\epsilon = 0$$

$$[x_0^2 + 4x_0 + 3] + [2x_0x_1 + 4x_1 + x_0 - 2]\epsilon + [x_1^2 + 2x_0x_2 + 4x_2 + x_1]\epsilon^2 = 0.$$

Now, if we equate the coefficient of each power of epsilon to zero, we can determine the approximate roots of our equation. In particular, notice that the coefficient $\epsilon^0 = 1$ corresponds to

$$x_0^2 + 4x_0 + 3 = 0 \implies x_0 = -3, \quad x_0 = -1.$$

The roots are the exact same roots that we arrive at by setting $\epsilon = 0$ in the exact solution or the original algebraic equation. We call these the *leading order* behaviour, since they are the roughest approximations we can make. If we wanted to improve our approximations, we could instead solve for x_1 by setting the coefficient of $\epsilon^1 = \epsilon$ to zero, giving

$$x_1 = \frac{2 - x_0}{2x_0 + 4} = \begin{cases} -\frac{5}{2}, & x_0 = -3 \\ \frac{3}{2}, & x_0 = -1. \end{cases}$$

Reconstructing our original roots, our approximations to the first order of ϵ are then

$$\begin{aligned} x &= -3 - \frac{5}{2}\epsilon \\ x &= -1 + \frac{3}{2}\epsilon. \end{aligned}$$

We can, of course, repeat this process several times to get even finer approximations, and eventually reconstruct our full solution in terms of an infinite series.

We can see how these approximations compare with our exact solution by considering the linear approximation to the discriminant,

$$\sqrt{(4 + \epsilon)^2 - 12 + 8\epsilon} = \sqrt{4 + 16\epsilon + \epsilon^2} = 2\sqrt{1 + 4\epsilon + \frac{\epsilon^2}{4}} \approx 2 + 4\epsilon.$$

If we plug this back into our exact solution, we find that

$$\begin{aligned} x_+(\epsilon) &\approx \frac{-(4 + \epsilon) + 2 + 4\epsilon}{2} = -1 + \frac{3}{2}\epsilon \\ x_-(\epsilon) &\approx \frac{-(4 + \epsilon) - 2 - 4\epsilon}{2} = -1 - \frac{5}{2}\epsilon \end{aligned}$$

which is the same as our approximate solution, confirming our result.

8.1.2 Differential Equations

Let us now consider an n -th order differential equation given by

$$\frac{d^n t}{dt^n} = f\left(t, y, \frac{dy}{dt}, \dots, \frac{d^{n-1}y}{dt^{n-1}}; \epsilon\right)$$

where $|\epsilon| \ll 1$. As before, we assume a solution of the form

$$y(t; \epsilon) = \sum_{k=0}^{\infty} y_k(t) \epsilon^k = y_0(t) + \epsilon y_1(t) + \epsilon^2 y_2(t) + \dots$$

to which we would substitute into our DE and solve for the coefficients, $y_k(t)$, in the series. It should be noted by the construction of this method that we would have to solve a differential equation for each $y_k(t)$. In many cases, they are linear, so it is possible to solve them exactly.

Example 8.1. Find the first three non-zero terms in a regular perturbation series solution for

$$\frac{dy}{dt} = y + \epsilon y^2, \quad y(0) = 1.$$

We assume a solution of the form

$$y(t; \epsilon) = y_0(t) + \epsilon y_1(t) + \epsilon^2 y_2(t) + \cdots$$

and evaluate the derivative,

$$\frac{dy(t; \epsilon)}{dt} = y'_0(t) + \epsilon y'_1(t) + \epsilon^2 y'_2(t) + \cdots.$$

At this point, we need only to substitute our solution into the differential equation to find that (we drop the arguments for clarity)

$$\begin{aligned} y'_0 + \epsilon y'_1 + \epsilon^2 y'_2 + \cdots &= y_0 + \epsilon y_1 + \epsilon^2 y_2 + \cdots + \epsilon(y_0 + \epsilon y_1 + \epsilon^2 y_2 + \cdots)^2 \\ &= y_0 + \epsilon y_1 + \epsilon^2 y_2 + \cdots + \epsilon[y_0^2 + 2\epsilon y_0 y_1 + \epsilon^2(y_1^2 + y_0 y_2) + \cdots] \\ &= y_0 + \epsilon(y_1 + y_0^2) + \epsilon^2(y_2 + 2y_0 y_1) + \cdots. \end{aligned}$$

Now, we must equate the coefficients of like powers of ϵ to each side, and so we find that

$$\begin{aligned} \epsilon^0 : \quad y'_0 &= y_0 \\ \epsilon^1 : \quad y'_1 &= y_1 + y_0^2 \\ \epsilon^2 : \quad y'_2 &= y_2 + 2y_0 y_1 \\ &\vdots \end{aligned}$$

Applying the initial condition, $y(0) = 1$, to the series, we find that

$$y_0(0) + \epsilon y_1(0) + \epsilon^2 y_2(0) + \cdots = 1.$$

For this initial condition to hold with our assumed solution form, it must be true that

$$y_0(0) = 1, \quad y_1(0) = 0, \quad y_2(0) = 0, \quad \cdots$$

which is a single initial condition for each coefficient of the series. We will solve for the coefficients, $y_k(t)$, beginning with the index $k = 0$, such that

$$y'_0 = y_0, \quad y_0(0) = 1.$$

This first differential equation is called the *leading order problem* of the differential equation, or the *order 1 problem*. The solution is simply

$$y_0(t) = e^t$$

and likewise it is called the *leading order solution* or *leading order behaviour* of the solution. Of course, for the index $k = 1$, the IVP is

$$y'_1 = y_1 + y_0^2, \quad y_1(0) = 0$$

and we can substitute our previous solution in to give

$$y'_1 - y_1 = e^{2t}.$$

This, now, is a linear DE with the integrating factor e^{-t} , to which we can solve and arrive at the solution

$$y_1(t) = e^{2t} + Ce^t.$$

Applying the IC, we find that

$$y_1(t) = e^t(e^t - 1)$$

and so we arrive at the new IVP for the index $k = 2$,

$$y_2' - y_2 = 2e^{3t} - 2e^{2t}, \quad y_2(0) = 0.$$

Solving this with the integrating factor e^{-t} , we arrive at

$$y_2(t) = e^t(e^t - 1)^2.$$

This lengthy process allows us to construct the full, perturbation approximation which is

$$y(t; \epsilon) \approx e^t + e^t(e^t - 1)\epsilon + e^t(e^t - 1)^2\epsilon^2.$$

An equation of this form is what we consider *weakly nonlinear*, since there is a small parameter multiplying the nonlinear terms. It should be noted that at step k in this example, we had to solve a linear differential equation for $y_k(t)$, which held for each equation that we had to solve. This is true, in general, when we solve weakly nonlinear DEs by perturbation theory. At the step, k , we solve a linear DE for the solution $y_k(t)$, and so the solution set becomes of the form $y_{kh}(t) + y_{kp}(t)$; only $y_{kp}(t)$ changes at each step, since the homogeneous solution remains consistent.

8.2 Duffing's Equation

Let us now consider a situation where we are unable to solve the given IVP. The model for the time evolution of an undamped pendulum is given by

$$ml \frac{d^2\theta}{dt^2} = -mg \sin(\theta), \quad \theta(0) = \theta_0, \quad \left. \frac{d\theta}{dt} \right|_{\theta=0} = 0$$

where l is the length of the pendulum, m is the mass of a bob, and g is the acceleration due to gravity. We let $\tilde{t}$ be the time passed and θ be the angle that the pendulum makes with the vertical, θ_0 being its corresponding initial angle. Applying a substitution to rewrite the equation in terms of dimensionless variables

$$u = \frac{\theta}{\theta_0} \quad \text{and} \quad t = \sqrt{\frac{g}{l}} \tilde{t}.$$

Thus, the differential equation becomes

$$\frac{d^2u}{dt^2} = -\frac{\sin(\theta_0 u)}{\theta_0}, \quad u(0) = 1, \quad \left. \frac{du}{dt} \right|_{t=0} = 0.$$

This new IVP is very nonlinear and depends on the parameter θ_0 . However, if we assume that the initial angle is small, that is, $|\theta_0| \ll 1$, then we can use perturbation methods to study the IVP. Let us rewrite the right side by Taylor expanding the sine function,

$$\frac{\sin(\theta_0 u)}{\theta_0} = u - \frac{\theta_0^2}{6} u^3 + \frac{\theta_0^4}{120} u^5 + \dots$$

and we will define $\epsilon = -\theta_0^2/6$. Dropping higher order terms, since θ_0 is assumed to be small, we arrive at

$$\frac{d^2 u}{dt^2} + u + \epsilon u^3 = 0, \quad u(0) = 1, \quad u'(0) = 0.$$

This new IVP is called *Duffing's Equation*, which is another weakly nonlinear differential equation. To solve it, we will look for a perturbation solution of the form

$$u(t, \epsilon) = u_0(t) + \epsilon u_1(t) + \dots$$

Substituting this into our DE (and dropping the arguments), we have

$$\begin{aligned} \frac{d^2}{dt^2}[u_0 + \epsilon u_1 + \dots] + [u_0 + \epsilon u_1 + \dots] + \epsilon[u_0 + \epsilon u_1 + \dots]^3 &= 0 \\ \left[\frac{d^2 u_0}{dt^2} + u_0\right] + \left[\frac{d^2 u_1}{dt^2} + u_1 + u_0^3\right]\epsilon + \mathcal{O}(\epsilon^2) &= 0. \end{aligned}$$

The initial conditions imply

$$\begin{aligned} u_0(0) + \epsilon u_1(0) + \dots &= 1 \\ \left.\frac{du_0}{dt}\right|_{t=0} + \epsilon \left.\frac{du_1}{dt}\right|_{t=0} + \dots &= 0 \end{aligned}$$

and looking at the leading order behaviour, we have the IVP

$$\frac{d^2 u_0}{dt^2} + u_0 = 0, \quad u_0(0) = 1, \quad u_0'(0) = 0.$$

The exact solution to the leading order behaviour including IC's is thus

$$u_0(t) = \cos(t).$$

Now, considering the case of ϵ^1 , the first order DE is

$$\frac{d^2 u_1}{dt^2} + u_1 = -u_0^3 = -\cos^3(t), \quad u_1(0) = 0, \quad u_1'(0) = 0.$$

Notice that this DE is only nonlinear with respect to u_0 , which suggests that the leading order solution, u_0 , forces the next order solution, u_1 , and appears as a non-homogeneous term in the equation. From the method of Laplace transforms, we can solve this DE,

$$\begin{aligned} (s^2 + 1)U_1 &= \mathcal{L}\{-\cos^3(t)\} \\ U_1 &= \mathcal{L}\{\sin(t)\}\mathcal{L}\{-\cos^3(t)\} \\ u_1 &= -[\sin(t) * \cos^3(t)] \\ &= -\int_0^t \sin(t-\tau) \cos^3(\tau) d\tau \\ &= -\frac{1}{4} \int_0^t \sin(t-\tau) [\cos(3\tau) + 3\cos(\tau)] d\tau \end{aligned}$$

where we have used the identity

$$\cos^3(t) = \operatorname{Re}\left(\frac{e^{it} + e^{-it}}{2}\right)^3 = \operatorname{Re}\left(\frac{e^{3it} + e^{-3it}}{8} + \frac{3e^{it} + 3e^{-it}}{8}\right) = \frac{1}{4}[\cos(3t) + 3\cos(t)]$$

Using the sine difference identity, the integral can thus be rewritten into

$$\begin{aligned}
 u_1 &= -\frac{1}{8} \int_0^t [\sin(t+2\tau) + \sin(t-4\tau)] d\tau - \frac{3}{8} \int_0^t [\sin(t) + \sin(t-2\tau)] d\tau \\
 &= -\frac{1}{8} \left[-\frac{1}{2} \cos(t+2\tau) + \frac{1}{4} \cos(t-4\tau) \right]_0^t - \frac{3}{8} \left[\tau \sin(t) + \frac{1}{2} \cos(t-2\tau) \right]_0^t \\
 &= \frac{1}{32} \cos(3t) - \frac{1}{32} \cos(t) - \frac{3}{8} t \sin(t) \\
 &= \frac{1}{32} [\cos(3t) - \cos(t) - 12t \sin(t)].
 \end{aligned}$$

Thus, the perturbative solution to Duffing's Equation is of the form

$$u(t) = \cos(t) = \frac{1}{32} [\cos(3t) - \cos(t) - 12t \sin(t)]\epsilon + \mathcal{O}(\epsilon^2).$$

There is, however, a problem in this solution. The perturbation expansion we use attempts to find a solution where the first term is always larger than the second. That is, $u_0 \gg \epsilon u_1$. Here, there are terms in $u_1 \propto t$ and not the sine or cosine of t ; we cannot expect the solution to be valid for all time since $u_1/u_0 \rightarrow \infty$ as $t \rightarrow \infty$.

In particular, the approximated solution fails for $t = \mathcal{O}(\epsilon^{-1})$, so we expect the interval of time in which the solution is valid depends on the value of ϵ . Isolating the problematic term, which is

$$t \sin(t),$$

we say that it is a *secular term*. It appears because the forcing term in the u_1 equation is resonant since the natural frequencies of the leading order and the next order terms are the same.

8.3 The Poincaré-Lindstedt Method

The nonlinear pendulum is just a single case of a secular term causing problems in the perturbative solution. In general, if a nonlinear differential equation has periodic solutions, then the period may depend on the initial conditions of the system. In this way, the rescaling of time, $t = \sqrt{g/l}\tilde{t}$ is not particularly appropriate, leading to the invalidity of the perturbative approximation for large time.

To fix this, we can assume the period is dependent on the initial position. That is, the rescaling of time can be change to

$$\tau = \sqrt{\frac{g}{l}} [1 + \Omega(\theta_0)] \tilde{t}$$

where we have assumed the leading order frequency is $\sqrt{g/l}$. For Duffing's equation, we can incorporate τ by making a perturbative approximation to the time, t ,

$$t = (1 + \epsilon\omega_1 + \epsilon^2\omega_2 + \dots)\tau.$$

The first term is one, since the frequency of Duffing's equation at the index $\epsilon = 0$ is also 1. We will choose ω_1 and ω_2 to eliminate our secular terms in the original perturbative expansion.

However, we must transform Duffing's equation from t to τ by the chain rule, where

$$\begin{aligned}
 \frac{d}{d\tau} &= \frac{dt}{d\tau} \frac{d}{dt} = (1 + \epsilon\omega_1 + \epsilon^2\omega_2 + \dots) \frac{d}{dt} \\
 \frac{d^2}{d\tau^2} &= \left[\frac{dt}{d\tau} \right]^2 \frac{d^2}{dt^2} = (1 + \epsilon\omega_1^2 + \epsilon^2\omega_2 + \dots)^2 \frac{d^2}{dt^2}.
 \end{aligned}$$

Using this transformation, Duffing's equation becomes

$$\begin{aligned}\frac{d^2 u}{d\tau^2} + (1 + \epsilon\omega_1 + \epsilon^2\omega_2 + \cdots)^2(u + \epsilon u^3) &= 0 \\ \frac{d^2 u}{d\tau^2} + [1 + 2\epsilon\omega_1 + \epsilon^2(\omega_1^2 + 2\omega_2) + \cdots](u + \epsilon u^3) &= 0.\end{aligned}$$

Substituting a perturbation expansion for $u(\tau)$ into this equation, such that

$$u = u_0(\tau) + \epsilon u_1(\tau) + \epsilon^2 u_2(\tau) + \mathcal{O}(\epsilon^3)$$

we find the equation

$$\begin{aligned}\frac{d^2}{d\tau^2}[u_0 + \epsilon u_1 + \epsilon^2 u_2 + \mathcal{O}(\epsilon^3)] \\ + [1 + 2\epsilon\omega_1 + \epsilon^2(\omega_1^2 + 2\omega_2) + \mathcal{O}(\epsilon^3)]\{u_0 + \epsilon u_1 + \epsilon^2 u_2 + \mathcal{O}(\epsilon^3) + \epsilon[u_0^3 + 3\epsilon u_0^2 u_1 + \mathcal{O}(\epsilon^2)]\} &= 0.\end{aligned}$$

Each equation for the coefficients in ϵ are then

$$\begin{aligned}\epsilon^0 : \quad \frac{d^2 u_0}{d\tau^2} + u_0 &= 0 \\ \epsilon^1 : \quad \frac{d^2 u_1}{d\tau^2} + u_1 &= -u_0^3 - 2\omega_1 u_0 \\ \epsilon^2 : \quad \frac{d^2 u_2}{d\tau^2} + u_2 &= -3u_0^2 u_1 - 2\omega_1(u_1 + u_0^3) - (\omega_1^2 + 2\omega_2)u_0\end{aligned}$$

with the IC's

$$\begin{aligned}\epsilon^0 : \quad u_0(0) &= 1, \quad \left. \frac{du_0}{d\tau} \right|_{\tau=0} = 0 \\ \epsilon^1 : \quad u_1(0) &= 0, \quad \left. \frac{du_1}{d\tau} \right|_{\tau=0} = 0 \\ \epsilon^2 : \quad u_2(0) &= 0, \quad \left. \frac{du_2}{d\tau} \right|_{\tau=0} = 0.\end{aligned}$$

Here, the leading order solution is the same as before, which is

$$u_0(0) = \cos(\tau).$$

The first order solution is given by the DE

$$\frac{d^2 u_1}{d\tau^2} + u_1 = -u_0^3 - 2\omega_1 u_0 = -\cos^3(\tau) - 2\omega_1 \cos(\tau) = \left(-2\omega_1 - \frac{3}{4}\right)\cos(\tau) - \frac{1}{4}\cos(3\tau).$$

The first term in $\cos(\tau)$ has the same frequency as the natural frequency on the left hand side. We expect it to thus produce terms involving

$$\tau \sin(\tau) \quad \text{and} \quad \tau \cos(\tau)$$

in the particular solution, so it is secular. We do *not* want secular terms, so we will thus choose $\omega_1 = -3/8$, resulting in the equation

$$\frac{d^2 u_1}{d\tau^2} + u_1 = -\frac{1}{4}\cos(3\tau).$$

Here, the homogeneous solution is

$$u_{1h} = A \cos(\tau) + B \sin(\tau)$$

and the particular solution is of the form

$$u_{1p} = C \cos(3\tau).$$

The constant can be determined trivially, and turns out to be $C = 1/32$, so the first order solution is

$$u_1 = A \cos(\tau) + B \sin(\tau) + \frac{1}{32} \cos(3\tau).$$

The ICs imply that $A = -1/32$ and $B = 0$, such that

$$u_1 = \frac{1}{32} [\cos(3\tau) - \cos(\tau)].$$

Thus, the perturbative solution up to first order is of the form

$$u(\tau) = \cos(\tau) + \frac{\epsilon}{32} [\cos(3\tau) - \cos(\tau)], \quad t = \left(1 - \frac{3}{8}\epsilon\right)\tau.$$

We can transform back to the original timescale, which gives the solution in t ,

$$u(t) = \cos\left(\frac{8t}{8-3\epsilon}\right) + \frac{\epsilon}{32} \left[\cos\left(\frac{24t}{8-3\epsilon}\right) - \cos\left(\frac{8t}{8-3\epsilon}\right) \right].$$

Notice that this solution does not have any secular terms, and, as a result, has a constant amplitude. For smaller values of ϵ , the difference between the Poincaré-Lindstedt method and the origin perturbative method becomes apparent only when $t \sim 1/\epsilon$.

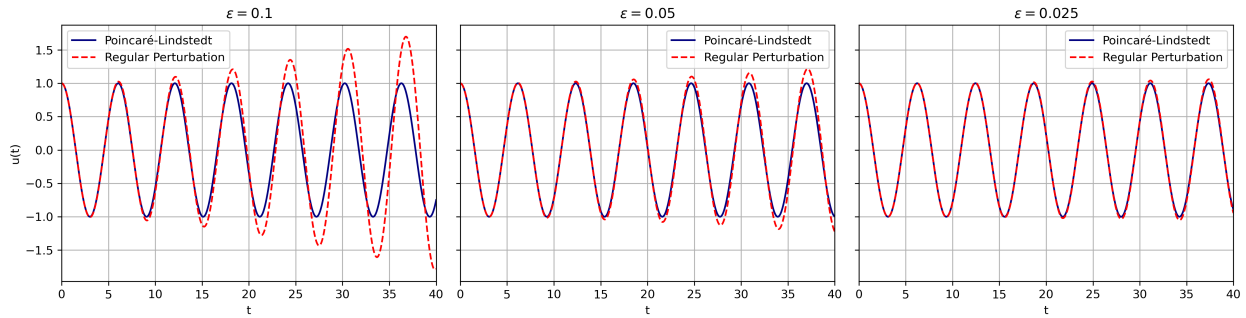


Figure 13: Approximate solutions to Duffing's equation using the regular perturbation expansion and the Poincaré-Lindstedt method.

Of course, we could continue to find the second order solution by substituting in the value of ω_1 and the first two terms of our approximate solutions into

$$\frac{d^2 u_2}{d\tau^2} + u_2 = -\frac{3}{32} \cos^2(\tau) [\cos(3\tau) - \cos(\tau)] - \frac{2\omega_1}{32} [\cos(3\tau) - \cos(\tau) + 32 \cos^3(\tau)] - (\omega_1^2 + 2\omega_2) \cos(\tau).$$

References